Exp No. 9 Mini Project - AI-Based Skill Gap and Career Recommendation System

AIM

To build an AI-based skill gap analysis and career recommendation web application that processes student skills, recommends suitable career paths using a machine learning classifier, identifies missing skills, and presents personalized guidance using a React frontend and API-based backend.

ALGORITHM

1. **Data Collection:** Obtain labeled career-path data and a structured catalog of job-role skills.
2. **Data Preprocessing:** Clean text, normalize skills and handle missing values.
3. **Career Label Mapping:** Map source career categories into supported job-role labels.
4. **Feature Extraction:** Convert skill descriptions into TF-IDF feature vectors.
5. **Train-Test Split:** Partition labeled samples into training and testing sets.
6. **Model Training:** Train and compare Logistic Regression and Linear SVM classification pipelines.
7. **Model Evaluation:** Assess model performance with accuracy, precision, recall, F1-score, and a confusion matrix.
8. **Model Serialization:** Save the selected trained career recommendation model using Joblib.
9. **Skill Gap Analysis:** Compare a student's current skills with required skills for suggested roles.
10. **Web Application:** Build a React-based frontend featuring student profiles, resume analysis, career matches, gap analysis and learning roadmaps.
11. **API Integration:** Connect the interface to backend services for predictions and career information.
12. **Deployment:** Run and validate the integrated application using a suitable development or deployment environment.

IMPLEMENTATION SCREENSHOTS

index.html

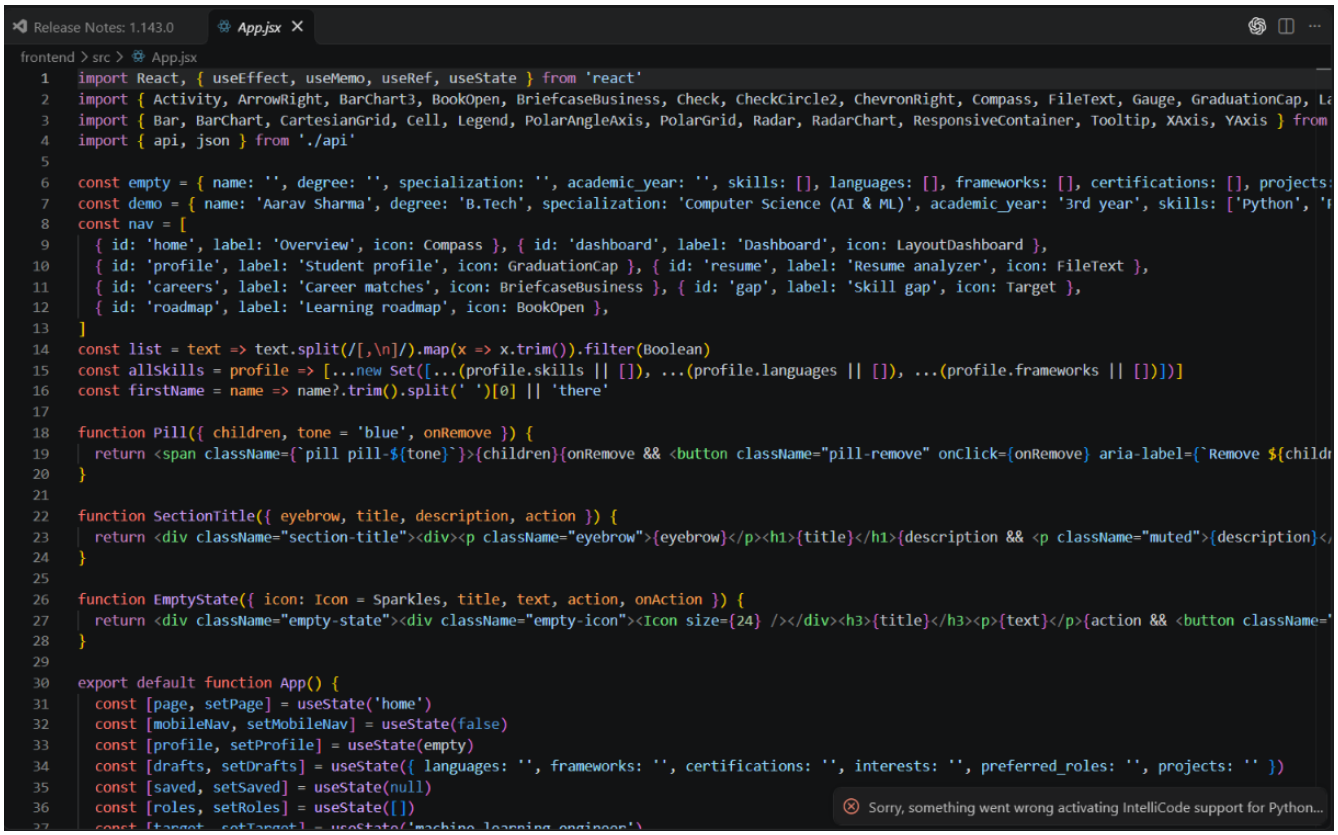


frontend/src/style.css



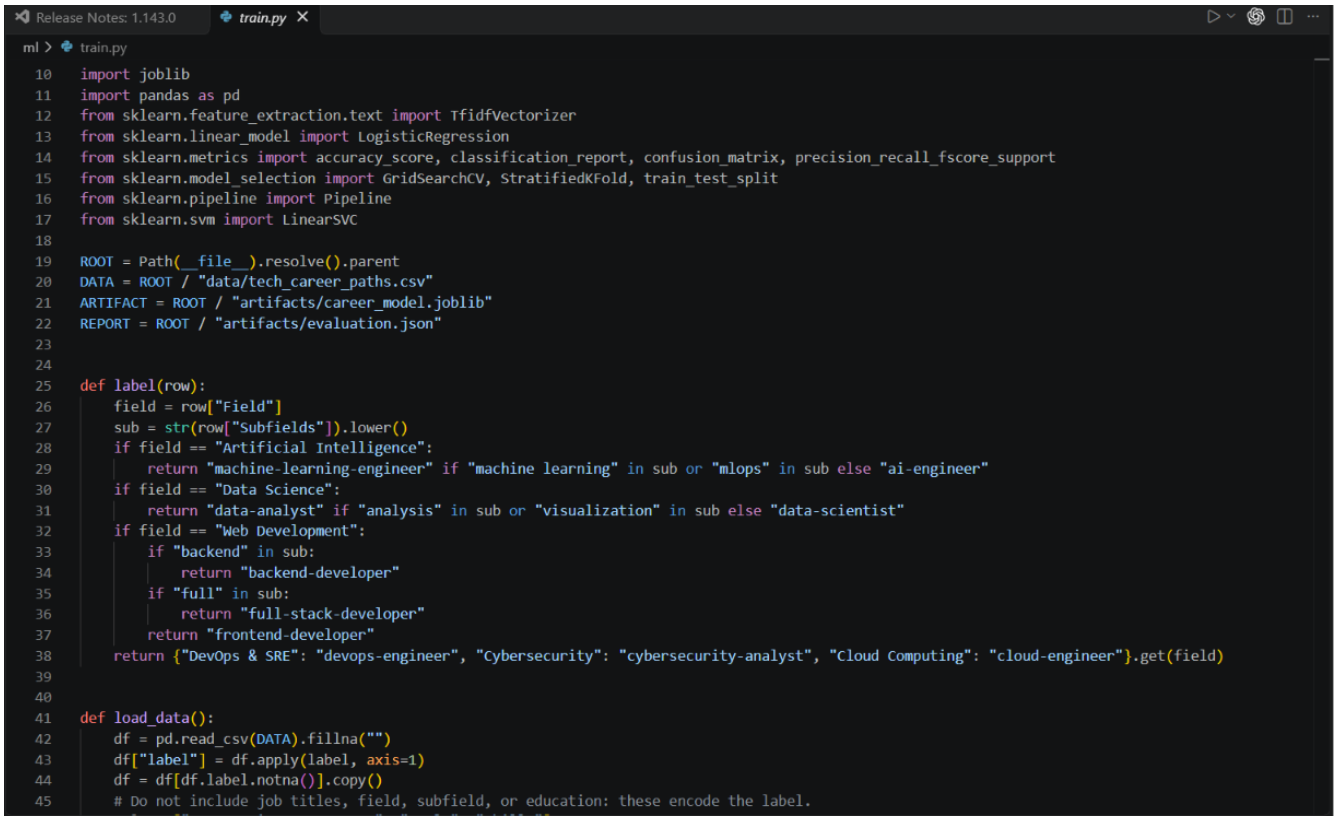
The React application is mounted at the root element. The stylesheet imports Tailwind CSS and defines the responsive interface styling.

frontend/src/App.jsx



The main React application defines interface navigation for overview, dashboard, student profile, resume analyzer, career matches, skill gap, and learning roadmap.

ml/train.py



```python
import joblib
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix, precision_recall_fscore_support
from sklearn.model_selection import GridSearchCV, StratifiedKFold, train_test_split
from sklearn.pipeline import Pipeline
from sklearn.svm import LinearSVC

ROOT = Path(__file__).resolve().parent
DATA = ROOT / "data/tech_career_paths.csv"
ARTIFACT = ROOT / "artifacts/career_model.joblib"
REPORT = ROOT / "artifacts/evaluation.json"


def label(row):
    field = row["Field"]
    sub = str(row["Subfields"]).lower()
    if field == "Artificial Intelligence":
        return "machine-learning-engineer" if "machine learning" in sub or "mlops" in sub else "ai-engineer"
    if field == "Data Science":
        return "data-analyst" if "analysis" in sub or "visualization" in sub else "data-scientist"
    if field == "Web Development":
        if "backend" in sub:
            return "backend-developer"
        if "full" in sub:
            return "full-stack-developer"
        return "frontend-developer"
    return {"DevOps & SRE": "devops-engineer", "Cybersecurity": "cybersecurity-analyst", "Cloud Computing": "cloud-engineer"}.get(field)


def load_data():
    df = pd.read_csv(DATA).fillna("")
    df["label"] = df.apply(label, axis=1)
    df = df[df.label.notna()].copy()
    # Do not include job titles, field, subfield, or education: these encode the label.
```

RESULT

The provided implementation screenshots show development of **SkillPath - AI Career Guidance**, including the React interface structure and a Python training script that uses TF-IDF features and classification models for career recommendations. The application is designed to support student skill analysis, recommended roles, skill-gap evaluation, and learning roadmaps.